

Kushagra Singh

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering **CGPA: 8.34/10**

Vellore, India

2023 – 2027

Professional Summary

Computer Science undergraduate with strong foundations in **Data Structures and Algorithms, distributed systems, machine learning, and backend development**. Experienced in building scalable software, GPU-accelerated applications, AI-powered systems, and secure computing solutions using modern software engineering practices.

Projects

Fog GPU – GPU-Accelerated Anomaly Detection Framework | *Python, CUDA, XGBoost*

2026

- Built a GPU-accelerated anomaly detection pipeline for fog computing environments to perform low-latency traffic filtering at edge nodes.
- Developed ensemble models using Isolation Forest and XGBoost, achieving **88% accuracy** and a **0.89 F1-score** on a 22k-sample evaluation dataset.
- Optimized inference throughput to **1.59 million samples/sec** using GPU-based stacked XGBoost for real-time processing of high-volume data streams.
- Benchmarked CPU and GPU execution by evaluating latency, throughput, and scalability across varying workloads.

EDDIE Learn Bot – AI Powered PDF Learning Platform | *React, LangChain, Ollama, Docker*

2025

- Developed a full-stack AI platform enabling users to upload PDF documents and interact with them through contextual conversations.
- Implemented Retrieval-Augmented Generation (RAG) using LangChain and Ollama to deliver document-aware responses with semantic search.
- Designed modular backend services and containerized the application with Docker for scalable deployment.
- Built an interactive interface supporting document viewing, AI-generated summaries, and context-aware question answering.

MSE Key - Entropy-Based Cryptographic Key Generator | *Python, OpenCV, Cryptography*

2025

- Designed a cryptographic key generation system using image-based physical entropy extracted from real-world digital media.
- Implemented OpenCV pipelines to identify high-entropy pixel regions for secure randomness generation.
- Generated cryptographic keys using hashing algorithms and Key Derivation Functions (KDFs).
- Evaluated generated keys using entropy and randomness metrics to verify cryptographic strength.

Technical Skills

Programming Languages: C/C++, Python, Java, Rust, SQL, JavaScript (ES6+)

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

Machine Learning: Scikit-learn, XGBoost, PyTorch, LangChain

GPU Computing: CUDA, Parallel Computing, GPU-Accelerated Inference

Backend & Infrastructure: Docker, Linux, Git, REST APIs, Microservices, Fog Computing

Databases: PostgreSQL, MongoDB

Security: AES, RSA, Cryptographic Hashing, Key Derivation Functions (KDFs), Wireshark, Nmap

Programming Profiles

LeetCode | **200+ Problems Solved**

leetcode.com/kushagra1424

Certifications & Achievements

FOSS United Hackathon | **Winner**

2026

IBM Cybersecurity Analyst Professional Certificate | **IBM**

2025